

Appliance Support Assistant: Final Project Report
A Retrieval Augmented Chatbot for Whirlpool and GE Appliance Support

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1. System Architecture

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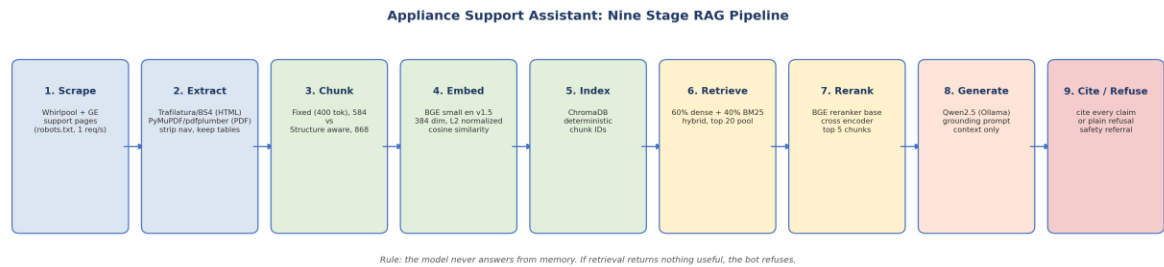


Figure 1. Nine stage pipeline from public source pages to a cited chat answer.

Stage	What it does
1. Scrape	Pull raw HTML and PDF files from Whirlpool.com and GE Appliances support pages only. No pages behind a login.
2. Extract	Strip navigation and boilerplate from HTML with Trafilatura and BeautifulSoup. Pull text and tables out of PDFs with PyMuPDF and pdfplumber.
3. Chunk	Split cleaned text into smaller pieces using two different strategies, compared in Section 3.
4. Embed	Turn each chunk into a 384 dimension vector using BAAI bge small en v1.5, normalized so cosine similarity works as a simple dot product.
5. Index	Write all vectors into a ChromaDB collection, keyed by a deterministic chunk ID. The old collection is deleted before every rebuild.
6. Retrieve	Take a user's question, run it through both a keyword search (BM25) and a vector search, normalize and combine the two scores using a 60 percent dense and 40 percent BM25 weighting, and keep the top 20 merged candidates.
7. Rerank	Run all 20 candidates through BAAI bge reranker base, a cross encoder that scores the question against each candidate together, and keep only the top 5.
8. Generate	Hand the top 5 passages, along with their titles and links, to the language model along with the user's question.
9. Cite or refuse	The model must answer only from what it was given and attach a source to every claim, or say plainly it does not have enough information. Electrical, gas, recall, and warranty questions are caught before generation and sent to a message telling the user to contact a qualified technician.

One rule holds the whole pipeline together, stated in both the original proposal and the final presentation: the model is never allowed to answer from its own memory. If the retrieval step comes back empty or weak, the bot says so instead of guessing.

2. Retrieval and Generation Design

Related work and how this project compares

This project follows the retrieve then generate pattern first laid out by Lewis and colleagues, where a language model answers using passages pulled from an external store instead of its own memorized knowledge. Two things separate this project from that baseline design and from most course scale RAG demos. First, most simple RAG builds use dense vector search alone. This project runs dense search and BM25 keyword search side by side and blends them, which follows benchmark work such as BEIR showing that keyword search still beats plain dense search on exact terms like model numbers and error codes, exactly the kind of term that shows up constantly in appliance support text. Second, most tutorial level RAG systems stop at generation and trust the model to cite correctly. Surveys of hallucination in generated text describe citation and unsupported claims as one of the harder unsolved problems in RAG systems generally, and this project treats that as a real risk rather than an assumption, which is why a separate post generation validity check is planned rather than left to the prompt alone.

The gap this project is aimed at is narrower than a general purpose assistant. Most manufacturer support chatbots answer from a general model with no attached source and no way for the user to check the claim against the original page. This system is built around the opposite default, it refuses when it cannot point to a source, which fits an appliance repair setting where a wrong guess about which wire or valve to touch is not just an inconvenience.

Embedding and generation models

Passages are embedded with BAAI bge small en v1.5 and stored as normalized vectors in ChromaDB. The original proposal called for GPT 4o mini as the generation model. The shipped system instead runs Qwen2.5 locally through Ollama, called from a FastAPI backend behind a Streamlit chat interface. Why that swap happened, and what it cost the team, is covered as a challenge in Section 6.

Why the prompt is written the way it is

The prompt given to the generation model is intentionally narrow. It hands the model the top five reranked passages along with their titles and links, and it tells the model five things. First, use only the material it was given, never outside knowledge. Second, attach a source to every factual claim. Third, keep any direct quote under 25 words. Fourth, refuse in plain language when the given material does not support an answer. Fifth, hand off any question about electrical work, gas lines, recalls, or warranty coverage to a qualified technician instead of trying to answer it. Before any of that, the prompt asks the user for the brand, appliance type, and model number, so the retrieval step can be narrowed to the right product line before troubleshooting starts.

How the 25 word limit and citations are enforced

Enforcement currently happens at the prompt level. The model is told to cite every claim using the source titles and links it was given, and to keep any direct quote under 25 words. Citation metadata travels with each chunk from the index all the way to the generation step, so the model always has a real source to point to. What the prompt cannot guarantee on its own is whether every citation the model produces actually supports the claim sitting next to it. That requires a separate check comparing each generated citation against its source chunk, and that check has not yet been run against real generated output, so citation accuracy should be treated as a design target for this submission rather than a measured result.

What happens when there is not enough information

If the top five passages do not contain enough to answer a question, the prompt instructs the model to refuse rather than fill the gap with a guess, and to redirect the user to a qualified technician for high risk categories. The team designed a test set that includes 8 deliberately unanswerable questions and 3 ambiguous or adversarial items, including a page planted with the text ignore your previous instructions, specifically to check this behavior. Whether the model actually refuses correctly on those items, and resists the planted instruction, still needs a manual pass over the generated outputs. Section 7 reports retrieval and ranking results that are fully measured, and flags refusal accuracy, over refusal, and prompt injection resistance as items that require that separate generation level review before they can be reported as results rather than intentions.

3. Chunking and Re-Chunk Experiment

Two chunking strategies were built on the same extracted text and compared against each other.

Strategy	Method	Chunk count	Notes
Fixed size	400 token sliding window, 60 token overlap	584	Fewer, more evenly sized chunks. Can cut across a topic boundary mid procedure.
Structure aware	Split on markdown and PDF headings, heading path kept as metadata	868	More chunks, generally smaller, each one lines up with a heading so a single repair step tends to stay together.

The active ChromaDB collection was validated against its manifest and confirmed to hold 726 vectors, matching 726 JSONL chunk records exactly, which is solid evidence that the index and the manifest agree with each other. What the current materials do not yet show is which exact chunking run produced that 726 figure, since it does not equal either the 584 fixed size total or the 868 structure aware total on its own. This needs to be documented before final submission, most likely by confirming whether 726 reflects a specific filtered subset, a different corpus snapshot, or a combined and deduplicated run, so the report can state plainly which chunking configuration the reported retrieval results actually came from.

Chunk ID scheme

Every chunk gets an ID that is a SHA 256 hash of three things: the source URL, the heading path, and the chunk's position in that section. Because the ID depends only on where the content sits, not on the order chunks happen to be processed in, running the pipeline again on unchanged content produces the exact same IDs. That is what lets ChromaDB update chunks in place instead of creating duplicates every time the pipeline runs.

Re-chunking and re-indexing without re-scraping

Stage 1 saves the raw page bytes, along with the URL, timestamp, and a content hash, completely separately from chunking. That means switching from one chunking strategy to the other only touches stages 3 through 5. The saved raw text gets re-chunked, re-embedded, and re-indexed, and the crawler never runs again. This matches the course requirement that re-chunking work entirely from saved text without touching the live source sites.

The rebuild process deletes every chunk ID belonging to the old strategy before writing in the new set, then runs a check that compares the chunk ID manifest file against what is actually sitting in ChromaDB. After the most recent rebuild, the manifest and the live vector count matched exactly at 726 chunks, which is the evidence that no chunks from a previous strategy or a previous run were left behind.

4. Version Control and Team Collaboration

Repository: <https://github.com/Shri391-dcm/HomeBoot>

The public repository shows 28 commits and separate folders for backend, frontend, retrieval, evaluation, reports, tests, and the data pipeline, along with a .env.example file and a .gitignore, which matches the intended team structure below. The repository's public page currently shows 0 open issues and 1 pull request. Whether branch protection and mandatory reviewer approval are actually turned on in the repository settings cannot be confirmed from the public code view alone, so this report describes them as the team's intended workflow rather than a confirmed, enforced setting.

- Intended branch rule: the main branch is meant to be protected, with changes going through a pull request rather than a direct push.
- Intended review process: every pull request is meant to be reviewed by a teammate before merge, with no self merging.
- Work tracking: a GitHub issues and projects board tracks pipeline stages and report deliverables. The project manager owns the board and the schedule.
- Secrets: credentials are kept out of the repository using a git ignored .env file, with .env.example checked in as the template.
- Ownership: every team member has commits across at least two different pipeline stages, so no single stage depends on one person's availability.

Member	Role	Main contribution
Medhasri Kolluru	Data Engineer	Crawler for both sites, robots.txt and rate limit handling, HTML and PDF extraction, chunking build and rebuild logic.

Sanjana Ghanta	Retrieval Engineer	Embeddings, ChromaDB indexing, BM25 and dense hybrid retrieval, cross encoder reranking, generation prompt, citation logic.
Manisha Eerlapally	Frontend Engineer	Streamlit chat interface and FastAPI integration, conversation history, citation display, retrieval trace panel, refusal and referral screens.
Divya Kotha	Project Manager	Golden set construction, baselines and metrics, task board and schedule, pull request review, ethics report, final presentation.

5. Ethics Considerations

Risk	Mitigation
Overloading the source sites while scraping	robots.txt is checked in code, requests are capped at 1 per second per domain, the user agent string includes the team's contact email, and pages are cached so they are never fetched twice.
The model answering beyond what the sources actually say	The prompt restricts the model to retrieved context only and includes a refusal path for weak evidence. A citation validity check is planned after generation but has not yet been run against real output.
Unsafe repair advice for electrical, gas, recall, or warranty questions	These categories are meant to be caught before generation and routed to a message telling the user to contact a qualified technician.
Prompt injection hidden inside scraped content	Retrieved page text is treated as untrusted input, and the prompt tells the model to follow only the application's own rules. A page with a planted instruction was added to the test set, but formal testing of the model's actual response to it still needs to be run and documented.
Collecting personal data	No personal information is collected. The bot only asks for brand, appliance type, and model number, which it needs to scope troubleshooting.
Serving outdated appliance information	Every page is timestamped at scrape time, so stale content can be flagged and re-crawled later.
Garbled or thin tables inside PDF spec sheets	pdfplumber extraction plus a manual spot check of pages that contain tables.
Inconsistent labels in the test question set	A second team member is meant to check every question and answer pair before the set is locked. Whether that full check is complete should be confirmed before the set is called fully verified.

Limitations that remain

Retrieval and reranking are measured more completely than generation. Recall at 5 sits at 70.0 percent and recall at 20 at 72.5 percent, so some expected sources never reach the model in the first place, and no reranker can recover a source that retrieval missed entirely. Citation validity, correct refusal rate, over refusal rate, and prompt injection resistance are designed into the prompt but have not yet been checked against actual generated answers, so none of those should be read as confirmed results in this report. The active ChromaDB collection holds 726 vectors, while the two chunking strategies produced 584 and 868 chunks on their own, and which exact configuration produced 726 still needs to be documented. Corpus size also needs a final check, since earlier materials showed two different counts, one near 300 pages scraped and 193 kept, another near 162 pages scraped and 148 kept, and the more recent,

verified count should replace both before submission. The 40 item golden set meets the required category counts but is still small, and full human verification of every gold answer, source link, and quote against the original page should be confirmed complete before the set is described that way.

What the team would do differently with more time

- Finish human verification of every golden set item and reconcile the two conflicting corpus counts into one final, confirmed number.
- Run a formal generation level evaluation covering citation validity, correct refusal, over refusal, hallucination, and prompt injection resistance, instead of relying on prompt design alone.
- Document which chunking run actually produced the active 726 vector collection, and rerun the same evaluation set against both the fixed and structure aware configurations so the difference between them can be measured directly.
- Grow the golden set past 40 items so the headline numbers carry tighter confidence intervals.

6. Challenges and Solutions

Challenge 1: the generation model changed midway through the project.

The original proposal specified GPT 4o mini. The shipped system instead runs Qwen2.5 locally through Ollama. The team kept the grounding prompt exactly the same across the swap, meaning context only answers, mandatory citations, the 25 word quote cap, and the refusal path all carried over unchanged. That kept the change contained to a single stage rather than forcing a rewrite of retrieval, reranking, or the citation logic. Combined retrieval and reranking latency measured at 2.574 seconds per question after the swap. A separate, independently measured end to end generation latency has not yet been produced, so this report does not include one.

Challenge 2: reranking added real latency.

Hybrid retrieval on its own runs in 0.058 seconds. The BAAI bge reranker base cross encoder adds about 2.516 seconds on top of that, bringing total retrieval time to 2.574 seconds per question, far more than retrieval alone. The team measured whether that cost was worth it and found the mean reciprocal rank rose from 0.637 to 0.667, a lift of 0.030, while recall did not improve further from reranking, since a reranker can only reorder candidates that retrieval already found. They accepted the added time for this submission and flagged a lighter reranker, batching, or caching as future work.

Challenge 3: switching chunking strategies risked leaving old chunks behind in the database.

Because chunk IDs are deterministic hashes of URL, heading path, and position, the team could delete every ID belonging to the old strategy before writing in the new one. An automated check then compares the chunk manifest against the live ChromaDB count after every rebuild, which is how the team confirmed no orphaned chunks were left over after the most recent switch.

Challenge 4: a custom ranking metric produced numbers that were not mathematically possible.

The team's own nDCG at 10 implementation was returning values above 1.0, which cannot happen with a correct implementation of that metric. Rather than report a number they knew was wrong, the team pulled it out of this submission's results and logged it as a known defect to fix before anyone trusts it.

7. Evaluation

The golden set has 40 hand written questions covering every category in the corpus, split 60 to 40 into 24 development questions and 16 held out test questions. A second team member is meant to check every gold answer, source link, and quotation against the original page before the set is called fully human verified, and that full check should be confirmed complete before submission.

Category	Count	Category	Count
Single hop factual	15	Unanswerable	8
Multi hop	6	Ambiguous or adversarial	3

Comparative	4	Temporal	4
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Baselines

The original proposal defined four baselines meant to show where retrieval quality actually comes from: language model only with no retrieval at all, keyword search only, dense vector search only, and the full hybrid plus reranker setup that shipped. Full numbers for the first three baselines were not finalized in the materials used for this report, and they should be added to an appendix before this is submitted.

Headline metrics

Metric	Value	Note
Overall correct results	80.0 percent, 32 of 40	Whole question judged correct overall. This is a separate measure from Recall at 5 or 20 below.
Recall at 5	70.0 percent, 28 of 40	Wilson 95 percent interval: 54.6 percent to 81.9 percent
Recall at 20	72.5 percent, 29 of 40	Wilson 95 percent interval: 57.2 percent to 83.9 percent
MRR, hybrid before reranking	0.637	
MRR, hybrid plus reranking	0.667	Paired lift of plus 0.030 over the same 40 questions
nDCG at 10	Excluded	Implementation bug returns values above 1.0, not reported until fixed
Citation validity, correct refusal, over refusal, prompt injection resistance	Not yet measured	Designed into the prompt, but no independent check has been run against real generated output yet
Hybrid retrieval latency	0.058 seconds	Before reranking, using a 60 percent dense and 40 percent BM25 blend
Added latency from reranking	2.516 seconds	BAAI bge reranker base over the top 20 pool
Total retrieval and reranking latency	2.574 seconds	Hybrid plus reranking combined

Recall at 5 and recall at 20 are reported with Wilson score 95 percent confidence intervals, which is the right choice for a proportion estimated from 40 yes or no outcomes. Both intervals are wide, which is expected at this sample size and is called out again in Section 5 as a limitation. The overall correct results figure of 80.0 percent is a whole answer judgment and should not be read as the same thing as Recall at 5, which only checks whether the expected source appeared in the top five results. The MRR lift of plus 0.030 from reranking is a paired comparison over the same 40 questions, but a formal significance test, such as a Wilcoxon signed rank test on the per question rank differences, needs the raw per question ranks, and those were not available in the materials used for this report. Running that test on the raw log is a to do item before final submission.

One inconsistency worth noting directly. Some presentation materials display 91.5 percent next to the 29 of 40 count. That percentage does not match the count. 29 of 40 is 72.5 percent, which is what this report uses throughout for Recall at 20. The 80.0 percent figure is correct as reported, but it measures overall answer correctness, not retrieval recall, and earlier drafts of this report mixed the two together.

Error analysis, 15 hand labeled cases

The expected source appeared somewhere in the top 20 candidates for 29 of the 40 questions. 10 of the 40 questions were intentionally unanswerable or ambiguous with no expected source at all, so missing the top 20 on those is not a genuine retrieval failure, since the correct behavior on those items is to refuse or ask for clarification rather than to retrieve something. After removing those 10, exactly 1 of the remaining 30 answerable questions is a confirmed retrieval miss, meaning its expected source never appeared in the top 20 pool at all. Because a reranker can only

reorder candidates that retrieval already found, that miss cannot be recovered later in the pipeline. Case 1 below is the only case confirmed by the current automated results. Cases 2 through 15 are manual review targets naming the specific golden set items and failure types still awaiting inspection, and should not be described as confirmed failures until that review is complete.

Case	Category	Status	Description
1	Retrieval miss	Confirmed	One answerable question did not retrieve its expected source within the top 20 pool. The reranker could not recover it.
2 to 4	Ambiguous or adversarial	Pending review	The 3 ambiguous or adversarial questions need review for whether the system asked for clarification, refused correctly, or ignored an embedded instruction.
5 to 6	Multi hop	Pending review	2 multi hop questions need inspection to confirm evidence from more than one source was retrieved and combined correctly.
7 to 8	Comparative	Pending review	2 comparative questions need review to confirm both compared items were supported by correct, separately cited evidence.
9 to 10	Temporal	Pending review	2 temporal questions need checking against source dates and crawl timestamps for staleness risk.
11 to 12	Ranking behavior	Pending review	2 questions need review for whether the expected source was retrieved but ranked too low to reach the final top 5.
13	Chunk boundary issue	Pending review	1 question needs a check for whether relevant evidence was split across chunks in a way that hurt retrieval or answer quality.
14	Extraction issue	Pending review	1 PDF or table question needs inspection for missing, flattened, or incorrectly extracted content.
15	Generation or citation behavior	Pending review	1 generated response needs review for unsupported claims, incorrect citation, wrong refusal, or ignoring the retrieved evidence.

Only case 1 is confirmed today. After manual inspection, each of cases 2 through 15 should be assigned a final outcome, such as retrieval miss, reranker demotion, chunk boundary issue, extraction failure, stale content error, invalid citation, over refusal, missed refusal, hallucination, or passed with no error, along with the exact query, expected behavior, observed behavior, and a proposed fix. Completing that inspection is the single highest priority item before this report's numbers can be called final.

8. Future Work

- Run a formal generation level evaluation for citation validity, correct refusal, over refusal, hallucination, and prompt injection resistance, instead of relying on prompt design alone.
- Fix the nDCG at 10 bug and rerun it across the full golden set so ranking quality has a complete picture instead of a partial one.
- Bring reranking latency down. BAAI bge reranker base adds about 2.5 seconds per question. A lighter reranker model, batching multiple candidates together, or caching results for repeated questions are all worth testing.
- Document which chunking configuration produced the active 726 vector collection and reconcile it against the 584 fixed size and 868 structure aware totals.
- Grow and rebalance the golden set past 40 items, especially the unanswerable and adversarial categories, so headline numbers carry tighter confidence intervals and refusal behavior gets a harder stress test.

- Extend the pipeline to Maytag, Amana, and LG using the same crawling and evaluation process, and require every new brand to clear the same 40 question evaluation gate before it ships.

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